Retail + Wholesale Business OS

Product Concept, Feature Blueprint & MVP Roadmap

A configurable, multi-tenant business management platform for retailers, wholesalers, distributors and industry-specific businesses - with an AI Business Copilot at its core.

Reference document - September 2026

1. Product Vision

The idea is to build a multi-business operating system that brings day-to-day business operations into one configurable platform. Instead of forcing every business into the same workflow, the platform adapts to the business through configurable industries, units of measurement, product attributes, pricing rules, document templates, permissions and workflows.

Core positioning: "One system that adapts to the business instead of forcing every business to work the same way."

The platform can serve a small retailer, a wholesale distributor, a textile company, an electronics seller, a grocery business, a hardware supplier, a furniture business or a service company using the same underlying system.

Long-term evolution

MVP -> SaaS business software -> AI Business Platform -> B2B Retailer/Wholesaler Network

2. Overall Product Structure

Area	Core capabilities
Products	Product catalogue, SKUs, barcodes, variants, descriptions, images, custom fields
Inventory	Stock, warehouses, transfers, adjustments, batches, low-stock alerts, valuation
People	Customers, suppliers, employees, roles, permissions, credit terms
Sales	Quotations, sales orders, delivery, invoices, payments, returns
Purchasing	Purchase requests, purchase orders, goods received, supplier invoices, returns
Documents	Invoices, receipts, payment vouchers, credit/debit notes, expense vouchers
Reports	Sales, inventory, purchasing, receivables, payables, profit and cash-flow views
POS	Barcode-based checkout, discounts, taxes, payment methods, receipts
AI Copilot	Onboarding, guidance, action execution, document extraction, insights
B2B Network	Long-term retailer-to-wholesaler discovery, pricing, orders and trading

The key data chain is: Product -> Purchase -> Inventory -> Sale -> Invoice -> Payment -> Profit. The value comes from keeping these relationships connected rather than treating each module as an isolated feature.

3. Dashboard and Daily Workspace

The home dashboard should answer "How is my business doing today?" before the user has to open any menu.

Dashboard signal	Example
Sales	AED 8,420 today
Purchases	AED 3,210 today
Outstanding	AED 12,850
Low-stock products	17

Invoices	24 created
Pending orders	8

Quick actions: Add Product, Create Invoice, Create Customer, Record Purchase, Add Stock, Create Quotation, Record Payment.

A visible AI panel can ask: "What would you like to do?" and accept natural-language requests.

4. Guided Business Setup

New users should not land in a complex ERP-style interface. Use an onboarding wizard that builds the business configuration first.

Step	What the user configures
Business	Business name, logo, address, country, currency, tax/VAT settings, business type
Industry	Retail, wholesale, distribution, manufacturing, textile, grocery, electronics, hardware, furniture, services or custom
Units	Default units and conversion rules
Products	Import or create starter products
People	Customers, suppliers and employees
Operations	Warehouses, pricing rules, invoice template and permissions

The industry selection should preconfigure sensible attributes and units without locking the user into them.

Industry-aware defaults

Industry	Useful defaults
Textile	Fabric type, GSM, color, width, pattern, roll, meter, yard, kg
Grocery	Piece, box, carton, kg, gram, liter, bottle, dozen, expiry
Electronics	Piece, box, set, pair, serial number, IMEI, warranty
Hardware	Material, diameter, thread size, grade, box, piece
Custom	User-defined fields, units and workflows

5. Product Management

Basic product data

- Product name
- SKU
- Barcode
- Category
- Brand
- Description
- Images
- Supplier
- Manufacturer

Pricing

- Purchase price
- Selling price
- Wholesale price
- Retail price
- Minimum selling price
- Customer-specific price lists
- Bulk pricing
- Promotional pricing

Inventory attributes

- Current stock
- Reserved stock
- Damaged stock
- Minimum/maximum stock
- Warehouse and location
- Shelf/rack
- Batch
- Expiry date

Variants

Shirt
- Black / S
- Black / M
- Black / L
- White / S
- White / M
- White / L

6. Flexible Units of Measurement (UOM) Engine

This is a core architectural requirement. Do not hard-code only piece/kg/liter. Build a reusable Unit of Measurement engine that supports standard units, business-specific units and conversion rules.

Family	Examples
Count	Piece, Unit, Pair, Set, Dozen, Gross
Weight	mg, g, kg, quintal, ton
Length	mm, cm, meter, km, inch, foot, yard
Area	sq mm, sq cm, sq m, sq ft, acre, hectare
Volume	ml, liter, cubic meter, gallon
Packaging	Box, Carton, Bag, Sack, Bundle, Roll, Pallet, Crate

Conversions

1 Carton = 24 Pieces
5 Cartons = 120 Pieces

1 Roll = 50 Meters
5 Rolls = 250 Meters

A product should be able to use different units for different operations. Example: Purchase unit = Roll, Inventory unit = Meter, Sales unit = Meter.

7. Sales Workflow

```
Customer -> Quotation -> Sales Order -> Delivery -> Invoice -> Payment
```

Small businesses can skip steps. Larger businesses can use the complete workflow, including warehouse picking and delivery.

8. Invoices and Document Customization

- Logo, colors, header and footer
- Bank details and tax information
- Payment terms
- Terms and conditions
- Signature
- QR code and barcode
- Custom fields
- Professional, minimal, retail, wholesale, tax and modern templates
- PDF export and customer delivery

9. Vouchers and Financial Documents

- Payment Voucher
- Receipt Voucher
- Purchase Voucher
- Sales Voucher
- Journal Voucher
- Credit Note
- Debit Note
- Discount Voucher
- Return Voucher
- Expense Voucher

```
Invoice #INV-1093
  -> Customer payment
  -> Receipt voucher
  -> Outstanding balance = 0
```

10. Purchasing

```
Supplier -> Purchase Request -> Purchase Order -> Goods Received -> Inventory Added -> Supplier Invoice -> Payment
```

- Supplier pricing
- Purchase history
- Partial deliveries
- Backorders

- Purchase returns
- Supplier credit
- Payment tracking
- Cost changes

11. Inventory Management

Opening stock
+ Purchases
+ Returns
+ Transfers
- Sales
- Damaged
- Adjustments
= Current stock

Multiple locations

- Warehouse A
- Warehouse B
- Retail Store
- Showroom
- Storage

Support stock adjustments, stock audits, transfers, damaged/expired goods, reserved stock, low-stock alerts and reorder levels.

12. Customer Management

Each customer should have a 360-degree view containing contact details, billing/shipping addresses, tax information, customer group, credit limit, payment terms, price list, purchase history and outstanding invoices.

13. Supplier Management

Supplier profiles should include supplied products, supplier pricing, delivery history, purchase history, payment history, outstanding amounts and average delivery time.

14. Multi-User Permissions

Role	Typical access
Owner	Full access
Manager	Operations and management; selected financial controls
Salesperson	Customers, sales and invoices
Warehouse employee	Inventory, receiving and transfers
Accountant	Invoices, payments and financial reports
Cashier	POS and payments

Use granular permissions: Can View, Can Create, Can Edit, Can Delete, Can Approve, Can Export.

15. Multi-Business / Multi-Branch Support

Business Group
- Retail Store
- Wholesale Division
- Warehouse
- Textile Business

Each business or branch can have its own products, inventory, customers, employees, invoices, taxes and warehouses, while owners can see consolidated performance.

16. AI Business Copilot

The AI should be more than a chatbot. It should understand the application, the business configuration and the available actions, while respecting permissions and requiring confirmation for sensitive operations.

Business Setup [done]
Tax Settings [done]
Units [done]
Products [35%]
Customers [0%]
Suppliers [0%]
Warehouse [0%]
Invoice Template [0%]

AI capabilities

- Explain features in simple language
- Guide users through screens and workflows
- Create drafts from natural-language requests
- Answer questions using business data
- Suggest configuration based on industry
- Surface warnings and insights
- Extract structured data from uploaded documents

17. Three Levels of AI Assistance

Level	Behavior	Example
Explain	Teaches a concept	"What is a sales order?"
Guide	Walks the user through the UI	"Create a sales order" -> highlights the correct path
Execute	Creates a draft or performs an allowed action	"Create an order for XYZ" -> prepares the order

Safe AI action model

The backend should expose controlled tools/actions to the AI. The AI should not have unrestricted direct database access. Sensitive actions such as finalizing invoices, deleting records, changing prices or issuing refunds can require explicit confirmation and role permission.

18. AI-Powered Business Insights

- Sales trend changes
- Supplier price increases
- Overdue receivables
- Products nearing stock-out
- Fast/slow-moving inventory
- Potential reordering opportunities
- Profitability questions in natural language

User: "Show me my top 10 most profitable products this year."
AI: Uses authorized business data -> calculates -> presents ranked results.

19. Reporting

Category	Example reports
Sales	Daily/monthly sales, product sales, customer sales, salesperson performance
Inventory	Stock valuation, stock movement, dead stock, fast-moving products, low stock, aging
Finance	Revenue, expenses, profit, receivables, payables, cash flow
Purchasing	Supplier spending, purchase history, price changes, supplier performance

20. POS

Search / Scan Product
Shirt AED 25
Jeans AED 80
Shoes AED 120

Subtotal | Discount | VAT | Total
Cash | Card | Other

Barcode scanner support should be designed into the product data model rather than added as an afterthought.

21. Wholesale-Specific Features

Feature	Example
Customer-specific pricing	Retail AED 30 | Wholesale AED 25 | VIP AED 22
Quantity pricing	1-9: AED 30 | 10-49: AED 27 | 50-99: AED 24 | 100+: AED 21
Credit sales	Invoice AED 15,000 | Paid AED 5,000 | Remaining AED 10,000 | Due in 30 days
Minimum order quantity	Minimum order: 50 pieces

22. Product History and Audit Trail

Important product information should be versioned rather than silently overwritten.

```
Sep 13  - Description changed
Sep 10  - Selling price changed: AED 28 -> AED 30
Sep 02  - Supplier changed
Aug 30  - SKU updated
```

Record who made the change, what changed, when it changed and, when useful, the previous value.

23. Document Automation

Allow a supplier invoice or other business document to be uploaded. AI can extract supplier, invoice number, line items and quantities, then propose a purchase or stock receipt for review.

```
Supplier: ABC Textiles
Invoice: INV-2038
25 x Cotton Roll
10 x Polyester Roll
50 x Thread Box
```

24. Notifications and Alerts

- Low stock
- Overdue invoices
- Supplier price changes
- Purchase order received
- Customer credit limit exceeded
- Stock transfer completed

Channels can include in-app notifications, email and later WhatsApp/SMS or push notifications.

25. Custom Fields and Industry Extensibility

Custom fields are essential if the platform is expected to serve different industries. Businesses should be able to create fields at appropriate levels such as product, customer, supplier, order or invoice.

Industry	Example custom fields
Textile	Fabric Type, GSM, Color, Width, Pattern
Electronics	Model, Voltage, Warranty, Serial Number
Hardware	Material, Diameter, Thread Size, Grade

26. Future B2B Marketplace / Network

Once many businesses use the operating system, the platform can connect retailers and wholesalers directly. This creates a second product layer on top of the ERP/SaaS foundation.

```
Retailer
  -> Search product
  -> Compare wholesalers, price, MOQ, availability
  -> Place order
  -> Wholesaler confirms
  -> Delivery
```

At that stage, the platform becomes a B2B commerce network as well as business management software.

27. Suggested Technical Architecture

Frontend: React / Next.js
Backend: FastAPI (Python)
Database: PostgreSQL
Cache / queues: Redis
File storage: S3-compatible object storage
Search: PostgreSQL initially; OpenSearch/Elasticsearch later if needed
AI: LLM + tool/function calling
Vector search: pgvector initially
Background jobs: Celery/RQ or equivalent

Keep the architecture modular so that AI, reporting, inventory and B2B commerce can evolve independently without turning the backend into one tightly coupled application.

28. MVP - What to Build First

Do not build the entire vision at once. A focused MVP can already be commercially meaningful.

1. Business setup and configuration
2. Users and roles
3. Products and variants
4. Units and unit conversions
5. Inventory
6. Customers
7. Suppliers
8. Sales and invoices
9. Purchases
10. Basic reports
11. AI onboarding assistant

Phase 2

POS, payments, advanced reports, notifications and workflow automation.

Phase 3

AI Business Copilot, document extraction, forecasting and richer business insights.

Phase 4

Retailer-to-wholesaler marketplace, supplier discovery, B2B ordering and network effects.

29. The Core UX Principle

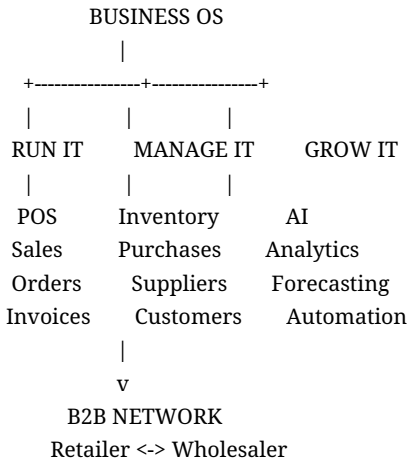
The product should not feel like "Here are 47 menus. Figure it out." It should feel like "Tell me what you are trying to accomplish, and I will help you do it."

User says	System behavior
"I received 20 cartons of shoes."	Recognize an inventory receipt, apply configured carton-to-

	piece conversion, prepare a stock receipt
"I want to buy fabric by roll and sell it by meter."	Configure purchase and sales units with a conversion rule
"Create an invoice for 50 shirts at AED 25 each."	Prepare an invoice draft for AED 1,250, then request confirmation before finalizing

30. Product Positioning and Vision

Positioning: A configurable business OS for retailers, wholesalers and distributors.



The strongest architectural choice is to make business configuration a first-class part of the system: industry, units, custom fields, workflows, permissions, document templates, pricing rules and tax settings. This is what lets one platform serve many industries without becoming rigid or unmanageable.

31. Quick Glossary

Term	Meaning
SaaS	Software as a Service - customers access the software online rather than installing and maintaining the whole system themselves.
UOM	Unit of Measurement - the system that defines pieces, meters, tons, cartons and conversions between units.
POS	Point of Sale - the checkout interface used by retailers to scan products, calculate totals and accept payment.
MVP	Minimum Viable Product - the smallest useful version to launch and validate before building the full vision.
AI Copilot	An AI assistant that explains, guides and eventually performs authorized actions inside the platform.
Multi-tenant	A SaaS architecture where multiple businesses use the same application while each business has isolated data and configuration.

32. Product Checklist for Future Planning

- Define the target first customer segment for the MVP.
- Define the business configuration model and tenant isolation rules.
- Design the UOM engine before building inventory screens.

- Design the product/variant model before building the sales flow.
- Define permission boundaries before connecting AI actions.
- Design invoices/documents as templates rather than fixed PDFs.
- Keep full audit history for financially important changes.
- Build onboarding as a guided setup experience, not a static tutorial.
- Validate the MVP with real retailer/wholesaler workflows before expanding into marketplace features.